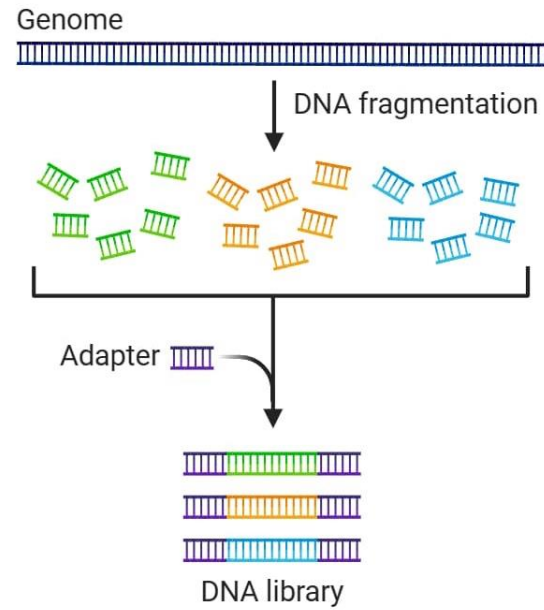


1. Quality control and trimming of raw short-read sequence data

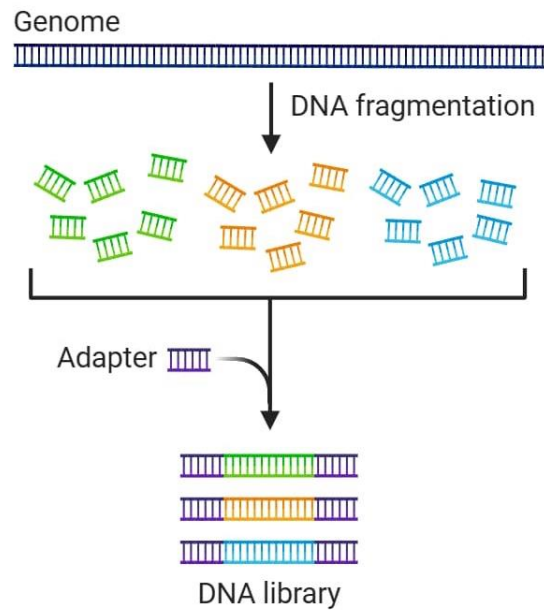
Illumina sequencing

① Library preparation



Illumina sequencing

① Library preparation



NovaSeq X series specifications

Output Range	~165 Gb - 16 Tb
Single reads per run	1.6 billion - 52 billion
Read length	2 × 150 bp
Run time	~13 hr - 48 hr

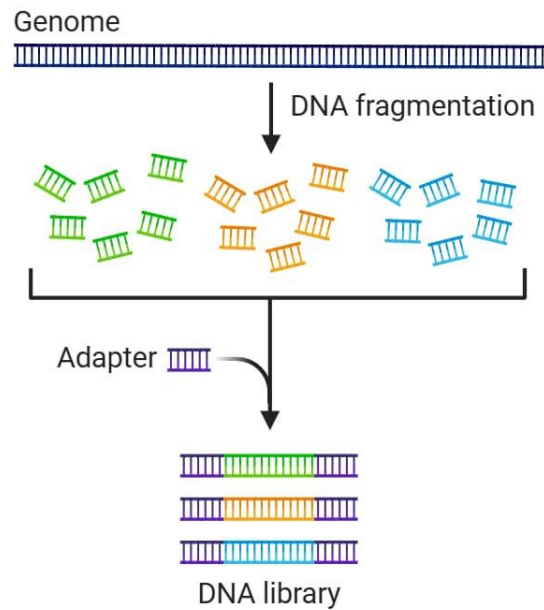
[View All NovaSeq X Specs](#)

 [View AR](#)

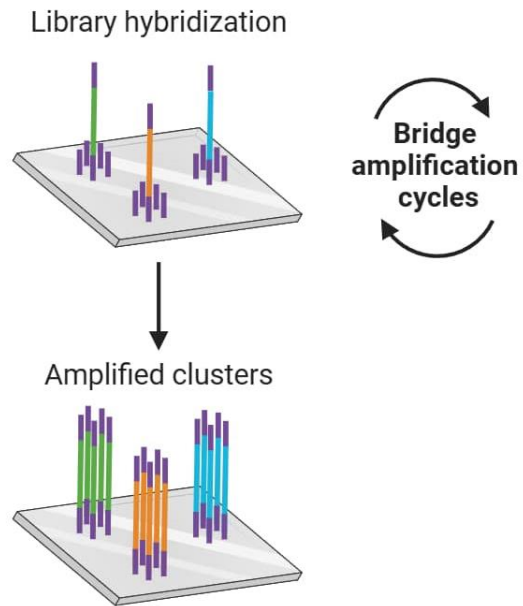
450 human genomes at 10x coverage per run

Illumina sequencing

① Library preparation

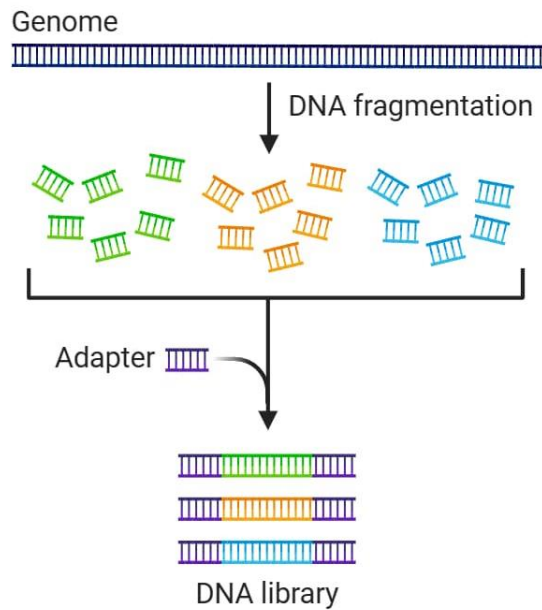


② DNA library bridge amplification

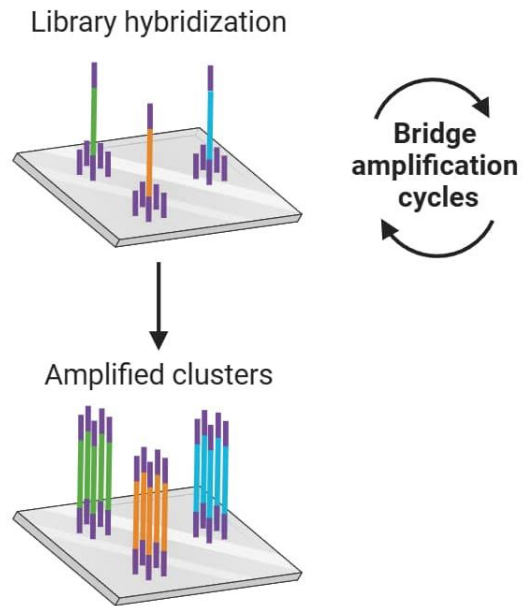


Illumina sequencing

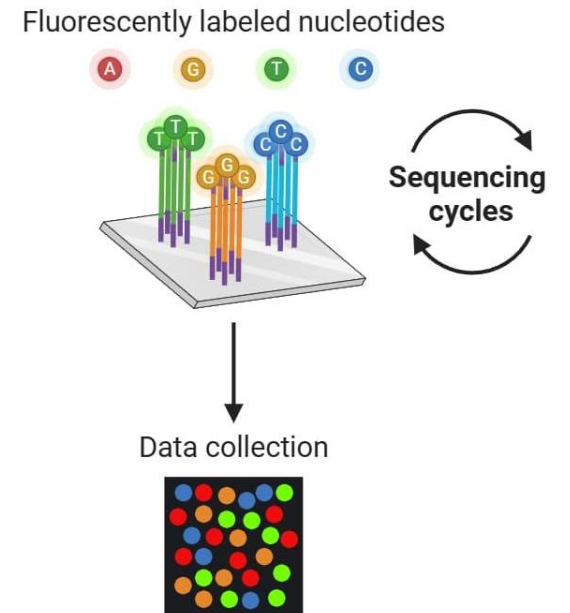
① Library preparation

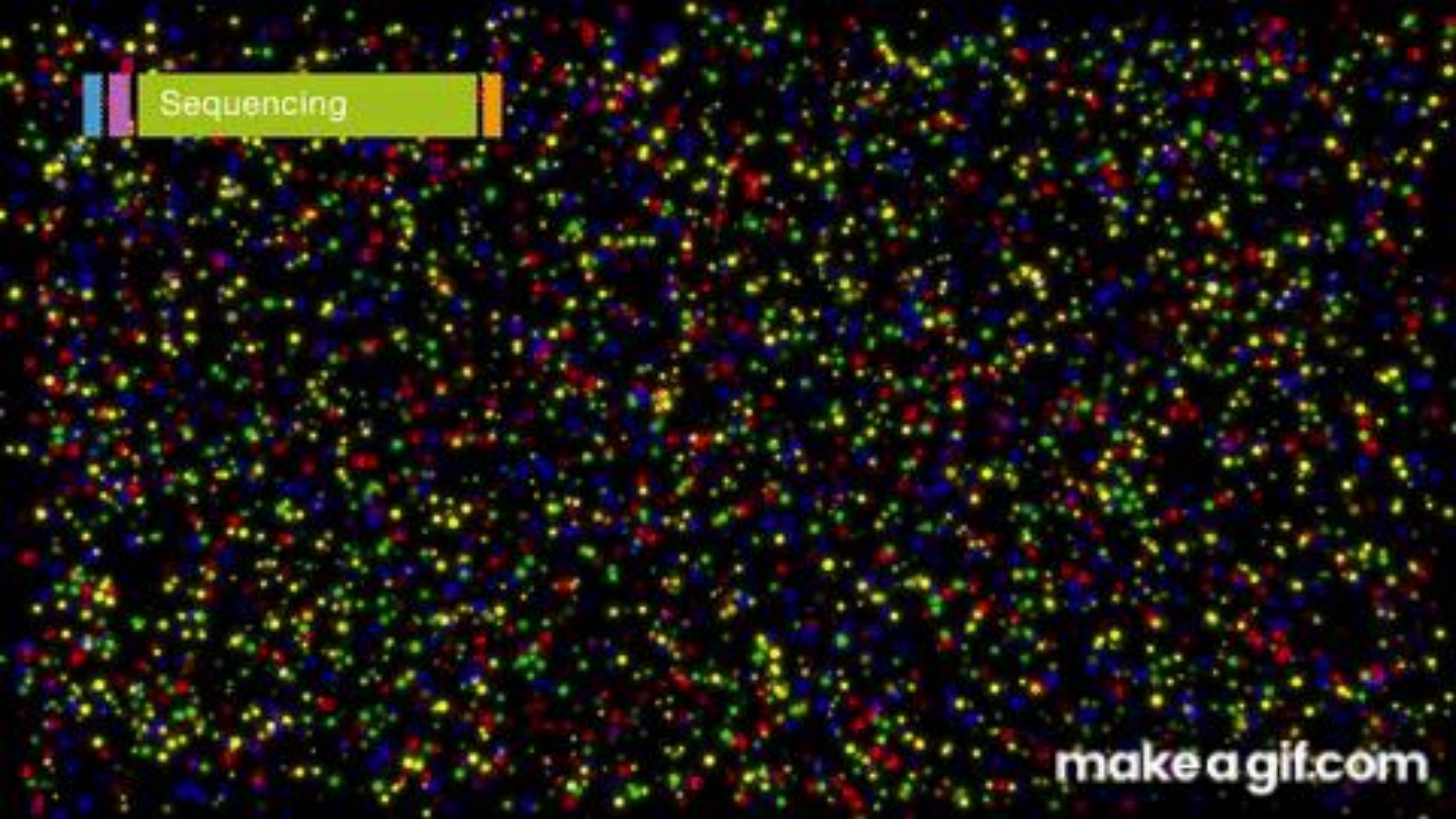


② DNA library bridge amplification



③ DNA library sequencing





Sequencing

makeagif.com

FASTQ Format

```
@A00180:10:H7NK5DMXX:1:1101:16821:1000 1:N:0:GGACTT+GTCGTTCG
TGCAGCAGCTAATGAGGAACCACTTCCTCCCTCCAGCCGCTCTAAATACCTCAGACAATAGGATCATCATAATAATCCCCTAGTCTGAACTG
+
8FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF
@A00180:10:H7NK5DMXX:1:1101:19090:1016 1:N:0:GGACTT+GTCGTTCG
TGCAGAAGAAGAAAGCACAAAGTATTTACGCCTATCCTTCATATTTTCCGCAAGGTA ACTATCTCGGTTTCATATCGAGATTTATATAGAATCT
+
FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF - FFFFFFFFFF
@A00180:10:H7NK5DMXX:1:1101:19325:1016 1:N:0:GGACTT+GTCGTTCG
TGCAGGAAGTTATGCAGGGGCATCCTGTATTATTAAATAGAGCACCTACTCTTCATAGATTAGGTATACAGGCGTTCCAACCTATTTTAGTGG
+
FF-88FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF - F - FFFFFFFF
@A00180:10:H7NK5DMXX:1:1101:30897:1016 1:N:0:GGACTT+GTCGTTCG
TGCAGAGTACATCAACAAAAGAAACCTAACTGCCCTACCGGCAAACCGGTAGAGTACCCTTCCCCAAAAGTATTACTCCCAGTCAATATAAGG
+
8F88FFF-FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF - FFFFFFFFFF - FFFFFFFFFF - FFFFFFFFFF - FFFFFFFFFF
```


FASTQ Format

```
@A00180:10:H7NK5DMXX:1:1101:16821:1000 1:N:0:GGACTT+GTCGTTCG
TGCAGCAGCTAATGAGGAACCACTTCCTCCCTCCAGCCGCTCTAAATACCTCAGACAATAGGATCATCATAATAATCCCCTAGTCTGAACTG
+
8FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF
@A00180:10:H7NK5DMXX:1:1101:19090:1016 1:N:0:GGACTT+GTCGTTCG
TGCAGAAGAAGAAAGCACAAAGTATTACGCCTATCCTTCATATTTCCGCAAGGTAAGTATCTCGGTTTCATATCGAGATTTATATAGAATCT
+
FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF - FFFFFFFFFF
@A00180:10:H7NK5DMXX:1:1101:19325:1016 1:N:0:GGACTT+GTCGTTCG
TGCAGGAAGTTATGCAGGGGCATCCTGTATTATTAAATAGAGCACCTACTCTTCATAGATTAGGTATACAGGCGTTCCAACCTATTTTAGTGG
+
FF-88F8FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF - F - FFFFFFFF
@A00180:10:H7NK5DMXX:1:1101:30897:1016 1:N:0:GGACTT+GTCGTTCG
TGCAGAGTACATCAACAAAAGAAACCTAACTGCCCTACCGGCAAACCGGTAGAGTACCCTTCCCCAAAAGTATTACTCCCAGTCAATATAAGG
+
8F888FFF-FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF - FFFFFFFFFF - FFFFFFFFFF - FFFFFFFFFF - FFFFFFFFFF
```

@HWI-ST227:389:C4WA2ACXX:7:1204:2272:59979

Header

GGAGGAAGGTCCTCGCTCCTCTTTCATATAAGGGAAATGGCTGAAT

Sequence

+

FFFFHHHHHHJIIJJJJJJJJIIJJJIGIGIGGIJJIIJIIJJJJJJIIII

Quality string

FASTQ Header

fastq header format (version > 1.8)

Sequence Header							+Sequence ID				
a	b	c	d	e	f	g	h	i	j	k	
@	HWI-ST486	:166	:C06K9ACXX	:7	:1101	:1443	:1995	1	N	0	:ACAGTG

a. unique instrument name

b. run id

c. flowcell id

d. flowcell lane

e. tile number within the flowcell lane

f. x-coordinate of the cluster within the tile

g. y-coordinate of the cluster within the tile

h. the member of a pair, 1 or 2 (paired-end or mate-pair reads only)

i. Y if the read fails filter (read is bad), N otherwise

j. 0 when no control bits are on

k. index sequence

FASTQ Sequence

- Contains the nucleotide sequence generated by the sequencer.
- Bases are normally represented by A, C, G and T.
- N indicates that the sequencer could not confidently identify the base.
- The sequence is the read as produced by the instrument and **may still contain adapters or other technical sequence.**

Adapter

```
GTGCCAGCCGCCGCGGTAAACGAAGGGACCTAGCGTAGTTCGGAATTACTGGGCTTAAAGAGTTCGTAGGTGGTTAAAAAGTTGGTGGTGAAATCCCAGAGCTTAACTCTGGAAGTGC
CATCAAACTTTTGTAGCTAGAGTATGATAGAGGAAAGCAGAATTTCTAGTGTAGAGGTGAAATTCGTAGATATTAGAAAGAATACCAATTGCGAAGGCAGCTTTCTGGATCATTACTGAC
ACTGAGGAAC
+
DDDDDIHHIIIIIIIIIIIIHHIIIIIIIIIIHHHHIIIIIIIIIIHHIIHFHHIIIIHHGHHIIIIIIIIIEHHIIHHIGHIIIFEHIIIIIEGHIII
IHFHHHHIIIIIIHII?GHHHGHIFHHHHIIHHIIIIHHIIIIIIHHIIGHGIIIIIIHHHHIIHFHHIIIIHHIGHGHHHHIIHHHHHHHHHHHH
HHIIIIIGFH
```

Phred+33 quality-scores encoding

Phred score	Q0	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Q11	Q12	Q13	Q14	Q15	Q16	Q17	Q18	Q19	Q20
Character	!	"	#	\$	%	&	'	(	)	*	+	,	-	.	/	0	1	2	3	4	5
Phred score	Q21	Q22	Q23	Q24	Q25	Q26	Q27	Q28	Q29	Q30	Q31	Q32	Q33	Q34	Q35	Q36	Q37	Q38	Q39	Q40	Q41
Character	6	7	8	9	:	;	<	=	>	?	@	A	B	C	D	E	F	G	H	I	J

Phred+33 quality-scores encoding

Phred score	Q0	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Q11	Q12	Q13	Q14	Q15	Q16	Q17	Q18	Q19	Q20
Character	!	"	#	\$	%	&	'	(	)	*	+	,	-	.	/	0	1	2	3	4	5
Phred score	Q21	Q22	Q23	Q24	Q25	Q26	Q27	Q28	Q29	Q30	Q31	Q32	Q33	Q34	Q35	Q36	Q37	Q38	Q39	Q40	Q41
Character	6	7	8	9	:	;	<	=	>	?	@	A	B	C	D	E	F	G	H	I	J

Phred score	Error probability
Q10	1 in 10
Q20	1 in 100
Q30	1 in 1,000
Q40	1 in 10,000

Phred+33 quality-scores encoding

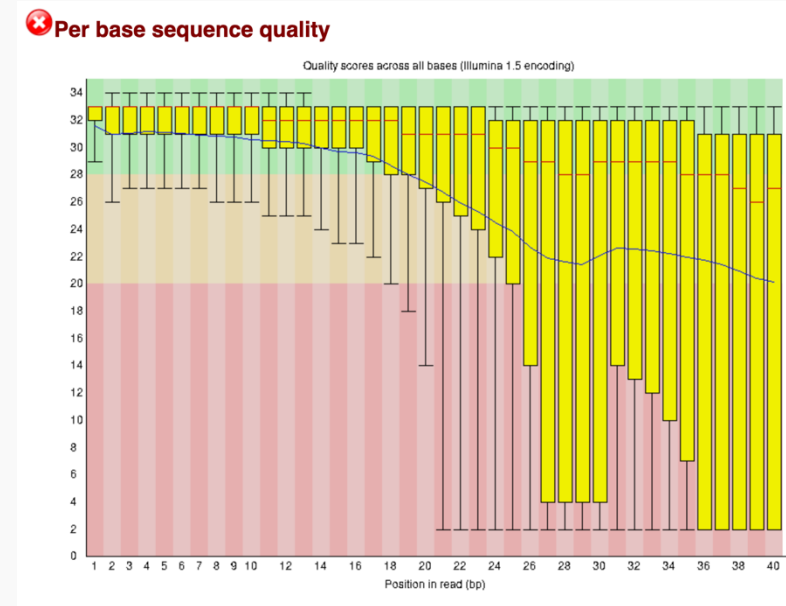
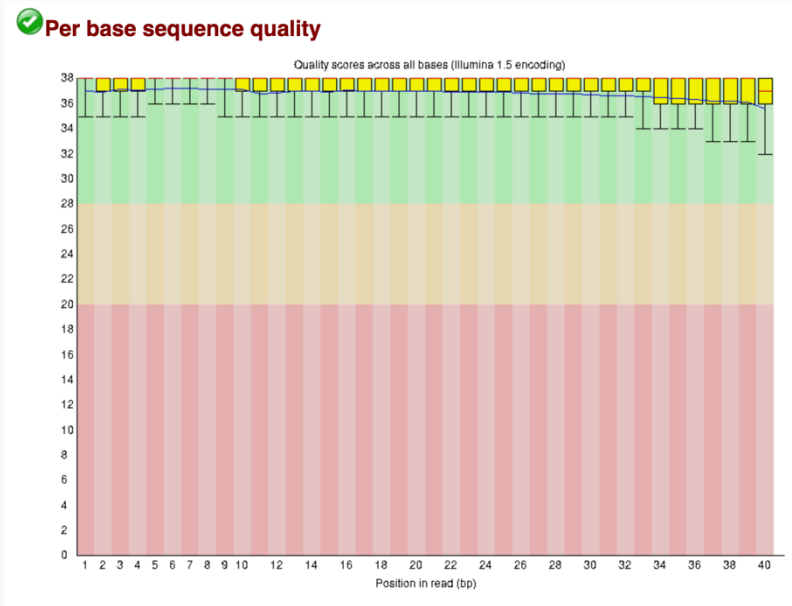
Phred score	Q0	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Q11	Q12	Q13	Q14	Q15	Q16	Q17	Q18	Q19	Q20
Character	!	"	#	\$	%	&	'	(	)	*	+	,	-	.	/	0	1	2	3	4	5
Phred score	Q21	Q22	Q23	Q24	Q25	Q26	Q27	Q28	Q29	Q30	Q31	Q32	Q33	Q34	Q35	Q36	Q37	Q38	Q39	Q40	Q41
Character	6	7	8	9	:	;	<	=	>	?	@	A	B	C	D	E	F	G	H	I	J

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```
@HWI-ST227:389:C4WA2ACXX:7:1204:2272:59979
GGAGGAAGGTCCTCGCTCCTCTTTCATATAAGGGAAATGGCTGAAT
+
FFFFHHHHHHHJIIJJJJJJJJIIJJJIGIGIGGIJJIIJIIJJJJJJIII
```


What do we need to do with these raw data?

- Preform initial QC
 - Check the length and quality of reads etc...



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NNNNNTTTTTTTTTTNGATCNNNNNAAAAAAAAANNNNNN
#####&&&&&&&&+++++00000+++++#####&&&&&

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 - Check the length and quality of reads etc...
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Let's practice